KEYSTONE: Readiness Audits for Dispute-Key Availability in Encrypted Rollups

Supplementary material

[Author details removed for peer review]

This supplement gives full proofs of the results that the article states with proof sketches (Lemmas 1–4, Corollary 2 and Theorems 2 and 3, with the limit after accepted audits), the stratified-sampling formula (Proposition S1), and the figures and tables moved out of the article for length. Notation follows the article; assumption, lemma, theorem and equation numbers refer to it. The temporal results are exact under the declared model; their parameters are synthetic.

1 Proofs for the audit (Lemmas 1–3)

Lemma (Lemma 1, correctness and knowledge soundness)**.** *An honest response passes verification. In the random-oracle model a verifying response satisfies $D_i = X^{x_i}$ except with negligible probability, and the proof is a proof of knowledge. Any t verified partials of C_1 with distinct indices I combine to PK_E^{ρ}.*

Proof. Completeness: $g^z = g^{\nu+cx_i} = a_1 Y_i^c$ and $X^z = a_2 D_i^c$ when $D_i = X^{x_i}$. The Chaum–Pedersen protocol is special-sound: two accepting transcripts with one commitment and different challenges $c \neq c'$ give $x_i = (z - z')/(c - c')$, and the same pair shows $D_i = X^{x_i}$. With the Fiat–Shamir hash as a random oracle, the forking lemma turns a party that produces verifying proofs for fresh canaries with non-negligible probability into an extractor that obtains x_i, and value soundness follows[1, 2]. Combination: $\prod_{i \in I} C_1^{x_i \lambda_i^I} = C_1^x = g^{\rho x} = PK_E^{\rho}$. □

Lemma (Lemma 2, simulatable audit transcripts)**.** *Model H_G and H as random oracles. A simulator that knows only g and $Y_1, \ldots, Y_n$ answers all oracle queries and produces honest custodians' audit responses for any audit context and sample. With at most N simulated proofs and Q oracle queries, simulated and real views are within statistical distance NQ/ℓ, so observing audits changes any adversary's advantage against* PAC *by at most $2NQ/\ell$.*

Proof. The simulator answers every fresh H_G query, from the protocol or the adversary, with g^a for a stored $a \leftarrow \mathbb{Z}_\ell^*$, which is exactly a random oracle onto $\mathbb{G} \setminus \{1\}$. For an honest custodian and canary $X = g^a$ it sets $D_i = Y_i^a = X^{x_i}$, the real response. Each proof comes from the honest-verifier zero-knowledge simulator: draw $c, z \leftarrow \mathbb{Z}_\ell$, set $a_1 = g^z Y_i^{-c}$ and $a_2 = X^z D_i^{-c}$, and program H at that input to c. Since a_1 is uniform, the input was queried before with probability at most Q/ℓ, and a union bound over N proofs gives the distance. Corrupted custodians' responses come from the adversary. Because H, H_G and the key-derivation function are independent and domain separated, this programming does not interfere with a reduction to the KEM that programs the key-derivation oracle, so an adversary against PAC that sees audits becomes one against the KEM alone, the probability of each output moving by at most NQ/ℓ. □

Lemma (Lemma 3, no production substitution)**.** *For a fixed C_1, a coordinator making Q queries to H_G finds an audit context whose canary equals C_1 with probability at most $Q/(\ell - 1)$.*

Proof. Each query returns an independent uniform element of $\mathbb{G} \setminus \{1\}$, equal to the fixed $C_1 \neq 1$ with probability $1/(\ell - 1)$; apply the union bound. □

Proposition (S1, stratified samples)**.** *Partition the committee into failure domains $G_1, \ldots, G_h$ of sizes n_j with k_j audit-ready custodians, and sample s_j custodians uniformly without replacement from each G_j, independently, with $\sum_j s_j = s$. Then*

$$\Pr[\mathsf{accept}] = \sum_{\substack{(i_1,\ldots,i_h) \\ i_1+\cdots+i_h \geq q}} \prod_{j=1}^{h} \frac{\binom{k_j}{i_j}\binom{n_j-k_j}{s_j-i_j}}{\binom{n_j}{s_j}}.$$

Proof. The per-domain counts of ready sampled custodians are independent hypergeometric (n_j, k_j, s_j) variables, and the audit accepts if and only if their sum is at least q. □

2 Proof of Lemma 4 (record-bound releases)

Lemma (Lemma 4)**.** *Let the dispute KEM be TDH2 with label $L_r = (E, \mathsf{id}_r, \text{policy hash})$, and let each custodian release $C_1^{x_i}$, with its DLEQ proof, only for a ciphertext whose TDH2 proof verifies and whose label is the authorized one. Under Assumptions 1 and 5 and static corruption of fewer than t custodians,* PAC^+ *holds for every record, and Lemma 2 continues to hold.*

Proof. Record identifiers are unique (Assumption 3), so distinct records have distinct labels. Fix a target record r and an adversary $\mathcal{A}$ against PAC^+. In *Game 0*, the PAC^+ experiment, $\mathcal{A}$ chooses two equal-length plaintexts m_0, m_1 for r; the challenger seals m_b for a uniform bit b; $\mathcal{A}$ sees the public key, the public shares, the corrupted custodians' shares, all audit transcripts, the sealed target and the releases of every other record, including records it commits itself, and outputs a guess b'. Let $\mathrm{Adv} = |\Pr[b' = b] - \frac{1}{2}|$.

Game 1 replaces the honest custodians' audit responses by the simulator of Lemma 2; by Lemma 2 and domain separation, $\Pr[b' = b]$ moves by at most NQ/ℓ.

Game 2 replaces the target's key-encryption key $\kappa_r = \mathrm{KDF}(PK_E^{\rho}, L_r)$ by a uniform key. A reduction $\mathcal{B}$ to Assumption 5 bounds the change. In the threshold model of Shoup and Gennaro[3], $\mathcal{B}$ receives the public key, the public shares and the corrupted shares, and forwards them. It obtains the challenge encapsulation for label L_r with a key κ^* that is either the real one or uniform, wraps a fresh data key K_r under κ^* with AES-GCM and associated data L_r, encrypts m_b under K_r, and gives $\mathcal{A}$ the sealed target. Audits are simulated as in Game 1, without the secret key. A release request for a record $r' \neq r$ concerns a ciphertext with label $L_{r'} \neq L_r$. If its TDH2 proof does not verify, honest custodians refuse it, as in the protocol. Otherwise it differs from the challenge at least in its label, so $\mathcal{B}$ obtains all decryption shares, with their proofs, from its oracle and computes the combined value by Lagrange interpolation, which is everything a release delivers. No release with label L_r is answered, because r is not released in the PAC^+ game, and the target's components under another label form a different ciphertext, valid only if its proof, which hashes the label, verifies; the chosen-ciphertext game permits such a query, and Assumption 5 is exactly the statement that no efficient adversary gains from it. $\mathcal{B}$ outputs 1 if $b' = b$, so the change is at most the advantage against Assumption 5.

Game 3 wraps a fixed all-zero key instead of K_r. Because κ^* is uniform and used once, the change is at most the advantage against AES-GCM as authenticated encryption. Now K_r is used only to encrypt m_b.

Game 4 encrypts m_0 instead of m_b under the uniform, once-used K_r; the change is at most the AES-GCM advantage, and in Game 4 the view is independent of b. Summing,

$$\mathrm{Adv} \leq \frac{NQ}{\ell} + \mathrm{Adv}^{\mathrm{cca}}_{\text{TDH2-KEM}} + 2\,\mathrm{Adv}^{\mathrm{ae}}_{\text{AES-GCM}},$$

which is negligible under Assumptions 1 and 5. The canary path does not use the KEM and its oracles are domain separated from the TDH2 oracles, so Lemma 2 holds unchanged. Proposition 1 shows that the label and validity checks are necessary: without them, the release of a re-submitted C_1 under a fresh identifier answers the challenge. □

3 Proof of Corollary 2 (stochastic deadline)

Lemma (Corollary 2)**.** *Suppose n_ℓ custodians can answer, each response is delivered independently with probability p_ℓ after a round trip with distribution function F, and the combiner finishes within B after the t-th valid response. Then* $\Pr[\mathrm{DDL}(r, E, \tau, \Delta_d)] \geq \Pr[\mathrm{Bin}(n_\ell, p_\ell F(\Delta_d - B)) \geq t]$.

Proof. Each of the n_ℓ responses is delivered within $\Delta_d - B$ of the request independently with probability $p_\ell F(\Delta_d - B)$, so their number M is $\mathrm{Bin}(n_\ell, p_\ell F(\Delta_d - B))$. On the event $M \geq t$, the t-th valid response arrives by $\tau + \Delta_d - B$ and the combiner finishes by $\tau + \Delta_d$, so DDL holds. □

In the R2b replication, F is the law of the injected two-leg delay, $p_\ell = (1 - \mathrm{loss})^2$, and B is the 99th percentile of the dispute completion time at zero injected delay, which absorbs transport, decoding and the combiner's work. B is therefore a high quantile rather than a bound, and the 0.99 target of Δ_{os} is approximate; the replication tests it on fresh traces.

4 Proofs for readiness over time

Model. Each of n custodians is an independent two-state continuous-time Markov chain with failure rate $\phi > 0$ and repair rate $\psi > 0$; $v = \psi/(\phi + \psi)$, and $p_\uparrow(u) = v + (1 - v)e^{-(\phi+\psi)u}$ and $p_\downarrow(u) = v(1 - e^{-(\phi+\psi)u})$ are the probabilities of being up at u given up, respectively down, at 0; note $p_\uparrow(u) - p_\downarrow(u) = e^{-(\phi+\psi)u} \geq 0$. In the domain-shock variant, h domains of $n_0 = n/h$ custodians each follow a further independent two-state chain with rates $\phi_d, \psi_d > 0$, and a custodian is ready when it and its domain are up. Faults are non-selective, audits at $T, 2T, \ldots$ draw fresh uniform samples and trigger no repair, and disputes arrive independently of the process. $A(k)$ is the acceptance probability of Eq. (1) of the article, non-decreasing in k.

Lemma (Lumped chains)**.** *(i) Without domain shocks, the number Z of ready custodians is a Markov chain with kernel*

$$P_u(k, k') = \sum_i \mathrm{b}(i; k, p_\uparrow(u))\, \mathrm{b}(k' - i; n - k, p_\downarrow(u))$$

and stationary law $\omega = \mathrm{Bin}(n, v)$. (ii) With domain shocks, the process $X = ((\delta_j, J_j))_{j=1}^h$, where δ_j indicates that domain j is up and J_j counts its members that are individually up, is a Markov chain whose kernel is the product of the per-domain kernels $P_u^\delta \otimes P_u^{(n_0)}$, and $Z = \sum_j \delta_j J_j$.

Proof. (i) The configuration process is a product of independent two-state chains. Started from a configuration with k custodians up, after time u each of them is up independently with probability $p_\uparrow(u)$ and each of the $n - k$ others with probability $p_\downarrow(u)$, so $Z(u)$ is the sum of independent $\mathrm{Bin}(k, p_\uparrow(u))$ and $\mathrm{Bin}(n-k, p_\downarrow(u))$ variables. This law depends on the configuration only through k, which is Dynkin's criterion for strong lumpability; hence Z is Markov with kernel P_u. Each custodian's stationary law is $(1 - v, v)$, and independence gives $\omega = \mathrm{Bin}(n, v)$. (ii) Domains are mutually independent, and inside a domain the domain chain is independent of the member chains; applying (i) inside each domain lumps its members to J_j, and lumping commutes with the product because the blocks are independent. A custodian is ready if and only if it and its domain are up, so $Z = \sum_j \delta_j J_j$. □

The pair (domain states, Z) is a function of X but not Markov in general: two configurations with the same domain states and the same Z whose down domain has different J_j have different futures once that domain recovers. Below, "the chain" is Z without shocks and X with shocks, ω its stationary law, and A, $\mathbf{1}_{<t}$ are read as functions of its state through Z.

Theorem (Theorem 2, risk given an audit history)**.** *Let the process be stationary before the first audit, and for outcomes $o_1, \dots, o_j$ let $\alpha_1 \propto \omega \odot e_{o_1}$ and $\alpha_i \propto (\alpha_{i-1} P_T) \odot e_{o_i}$, normalized, with $e_{\mathsf{acc}} = A$ and $e_{\mathsf{rej}} = 1 - A$. Then $\alpha_j(x) = \Pr[X(jT) = x \mid o_1, \dots, o_j]$, and for $0 \le u < T$, $\Pr[Z(jT + u) < t \mid o_1, \dots, o_j] = \alpha_j P_u \mathbf{1}_{<t}$.*

Proof. By the lemma, the chain sampled at the audit times is a Markov chain with kernel P_T. Given the path, the i-th sample is fresh, uniform and independent of everything else, and faults are non-selective, so o_i is conditionally independent of all other variables given the state at iT and is an acceptance with probability $A(Z(iT))$: the number of ready custodians in a uniform sample is hypergeometric and depends on the configuration only through Z. The audit history is therefore a hidden Markov model, and the recursion is Bayes' rule applied one audit at a time (the forward algorithm). For the dispute, the Markov property at jT and the independence of the dispute time give the law $\alpha_j P_u$ at $jT + u$; summing over states with $Z < t$ gives the claim. □

Proposition (Limit after accepted audits)**.** *If all audits accept, α_j converges geometrically, from any start, to the normalized left Perron vector α^* of $M = P_T \operatorname{diag}(A)$ restricted to the states S with $A > 0$.*

Proof. For $T > 0$, $0 < p_\downarrow(T) < v < p_\uparrow(T) < 1$, so each row of P_T is the convolution of two binomial laws with full supports $\{0, \dots, k\}$ and $\{0, \dots, n-k\}$, which has full support $\{0, \dots, n\}$; hence $P_T > 0$ entrywise. In the domain variant, each per-domain kernel is a Kronecker product of positive matrices and is positive, and so is their product. After one accepted audit, α is supported in S, and M restricted to $S \times S$ is strictly positive. By the Perron–Frobenius theorem, M has a simple dominant eigenvalue with a positive left eigenvector, and normalized iterates from any nonnegative nonzero start converge to it geometrically, by Birkhoff's contraction of the Hilbert projective metric. □

Theorem (Theorem 3, undetected failure)**.** *Let $U(T) = \frac{1}{T} \int_0^T (\omega \odot A) P_u \mathbf{1}_{<t} \, du$. With or without domain shocks, U is non-decreasing in $T > 0$ and $U(T) \ge U_0 = (\omega \odot A)\mathbf{1}_{<t}$, which is its limit as $T \to 0$. Without domain shocks,*

$$U(T) \le A(t-1)\,\omega(Z < t) + \sum_{k \ge t} \omega(k) A(k) \Pr[\mathrm{Bin}(k, p_\uparrow(T)) < t],$$

and the right-hand side is non-decreasing in T.

Proof. Interpretation. By stationarity, the independence of the dispute time and Theorem 2 with one audit, $U(T)$ is the long-run probability that a dispute fails and the most recent audit accepted.

Monotonicity and floor. Order the states coordinatewise (Z without shocks; each δ_j and J_j with shocks), so that A is non-decreasing and $h_0 = \mathbf{1}_{<t}$ is non-increasing; with shocks this holds because $Z = \sum_j \delta_j J_j$ is non-decreasing in every coordinate. The chain has three properties. (a) It is reversible with respect to ω: each coordinate is a two-state chain or a birth–death chain, both reversible, and a product of independent reversible chains is reversible. (b) Every transition changes one coordinate by one step. (c) P_u maps non-increasing functions to non-increasing functions: for a two-state chain this is $p_\uparrow(u) \ge p_\downarrow(u)$; for a birth–death count, raising the start by one replaces a $\mathrm{Bin}(1, p_\downarrow)$ summand by a $\mathrm{Bin}(1, p_\uparrow)$ one, a stochastically larger variable; and a product of independent monotone kernels is monotone. Let $\mathcal{L}$ be the

generator, $\langle f, h\rangle_\omega = \sum_x \omega(x) f(x) h(x)$, and $g(u) = (\omega \odot A) P_u h_0 = \langle A, P_u h_0\rangle_\omega$. Then $g'(u) = \langle A, \mathcal{L} P_u h_0\rangle_\omega = -\mathcal{E}(A, P_u h_0)$, where by (a)

$$\mathcal{E}(f, h) = \tfrac{1}{2} \sum_{x,y} \omega(x)\, \mathcal{L}(x, y)\, \big(f(y) - f(x)\big)\big(h(y) - h(x)\big)$$

is the Dirichlet form. By (b), each pair x, y with $\mathcal{L}(x, y) > 0$ is comparable; by (c), $P_u h_0$ is non-increasing; so each term has $(A(y) - A(x))((P_u h_0)(y) - (P_u h_0)(x)) \le 0$, and $g'(u) \ge 0$. Hence g is non-decreasing, its running mean $U(T)$ is non-decreasing in T, and $U(T) \ge g(0) = (\omega \odot A) h_0 = U_0$. As $T \to 0$, $P_u \to I$ and g is continuous, so $U(T) \to U_0$.

Bound. Without shocks, split U by the state $k = Z(0)$ at the audit. For $k < t$, bound $\Pr_k[Z(u) < t] \le 1$ and use $A(k) \le A(t-1)$. For $k \ge t$, let $N(u)$ count the k custodians up at 0 that are up at u. Then $Z(u) \ge N(u)$, since custodians down at 0 can only add, and $N(u) \sim \mathrm{Bin}(k, p_\uparrow(u))$, so $\Pr_k[Z(u) < t] \le \Pr[\mathrm{Bin}(k, p_\uparrow(u)) < t]$. Since $dp_\uparrow/du = -(1-v)(\phi + \psi) e^{-(\phi+\psi)u} < 0$ and the binomial law is stochastically increasing in its success probability, this is at most $\Pr[\mathrm{Bin}(k, p_\uparrow(T)) < t]$ for $u \le T$; averaging over u keeps the bound, and the same two facts make the right-hand side non-decreasing in T. □

Consequently, a target $\varepsilon < U_0$ is unattainable at every interval, and $T^*(\varepsilon) = \sup\{T : U(T) \le \varepsilon\}$ is well defined and found by bisection; the runner bisected $\log T$ over $[0.25\,\mathrm{h}, 30\,\mathrm{d}]$. The interval from the bound is conservative. Monotonicity of the posterior risk $R(T)$ is not proved; its intervals rest on a numerical check of 25 points per cell, which passed in every cell.

5 Additional figures and tables

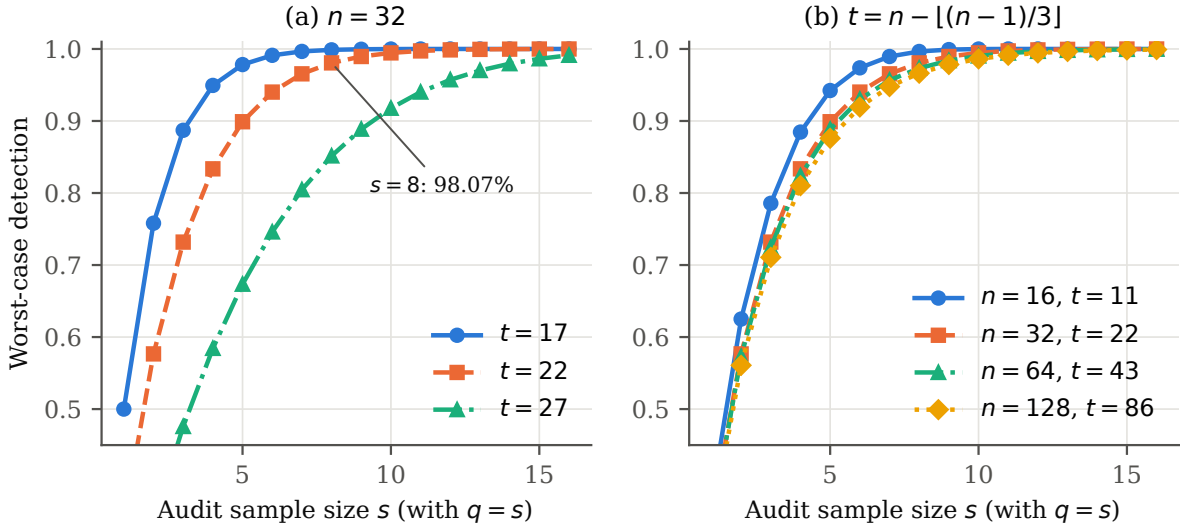


Figure S1: Exact worst-case detection $1 - \binom{t-1}{s} / \binom{n}{s}$ versus sample size with $q = s$. (a) Thresholds $t \in \{17, 22, 27\}$ at $n = 32$; the baseline $s = 8$, $t = 22$ is annotated. (b) Committee sizes 16–128 with $t = n - \lfloor (n-1)/3 \rfloor$. Values for $(n, s) \in \{(16, 4), (32, 8), (64, 12), (128, 16)\}$ are listed in Table S1.

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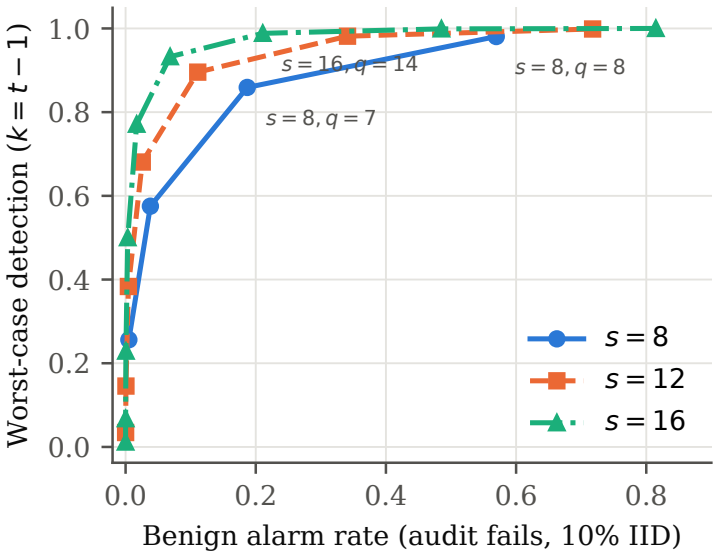


Figure S2: Detection versus benign alarm rate (the audit fails while the committee is healthy, $f = 0.10$) for each q, one curve per sample size. Exact values from Table 1 of the article and its full source grid.

Table S1: Scalability (R1 prototype, ffdhe groups) with $t = n - \lfloor (n-1)/3 \rfloor$. Detection is exact (Eq. (2) of the article, $q = s$). Audit CPU is s proofs plus s verifications; bytes are one request plus s responses. Dispute time is measured verification of t partials plus combination.

				Audit, ffdhe2048		Dispute: verify t + combine	
n	t	s	Detection	CPU	Bytes	ffdhe2048	ffdhe3072
16	11	4	88.46%	1063 ms	4,819	1585 ms	5040 ms
32	22	8	98.07%	2125 ms	9,415	3213 ms	10201 ms
64	43	12	99.66%	3188 ms	14,011	6230 ms	20086 ms
128	86	16	99.92%	4250 ms	18,607	12534 ms	40201 ms

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Table S2: Bulletin-board gas per operation, measured with Foundry 1.5.1 and Solidity 0.8.24 (test-body gas; fixture setup excluded), in the small harness committee named in the scope column; values are deterministic.

Bulletin-board operation	Gas	Harness scope
Register epoch	223,629	n=5, t=3; duplicate-identity validation enabled
Open audit	125,107	three sampled members; q=3
Submit audit response	57,882	first commitment from one sampled custodian
Mark invalid response	27,704	admin records a non-zero evidence hash
Record equivocation	51,710	second conflicting commitment from one custodian
Finalize audit	29,142	two responses for a three-member sample after deadline
Open dispute	144,149	n=5, t=3; all members eligible
Cancel request	5,709	admin cancels one open audit

Table S3: Mechanisms that check some form of availability or readiness. "Secret-key operation" means the check makes a custodian apply its secret share. Entries describe the mechanism class, not every system in it.

Mechanism	What is checked	Secret-key operation	Deadline evidence
Liveness heartbeat	A process answers	No	No
Threshold-wallet heartbeat (tBTC v2)[4]	The full signing group signs a heartbeat message	Yes (signing)	A failed heartbeat triggers fund migration
Proof of custody[5]	A validator holds the data it attests to	Secret-dependent bit	Challenge and penalty
PoR and PDP[6–8]	A server still stores a file	No	No
Data-availability sampling[9, 10]	Published data are retrievable	No	No
VSS, PVSS and DKG transcripts[11–13]	Shares were distributed consistently	At setup only	No
Threshold-release systems[14–16]	A release is performed when requested	At release only	System-specific
KEYSTONE	q of s sampled custodians can apply their shares now; t custodians serve a dispute by Δ_d	Before release (canary) and at release	Bulletin board

the 14th ACM Conference on Computer and Communications Security (CCS '07), pages 598–609. ACM, 2007. doi: 10.1145/1315245.1315318.

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Table S4: Readiness checks compared (R1 prototype). Costs are R1 medians for ffdhe2048; bytes are canonical request plus response transcripts.

Readiness check	Exercises the secret share?	Touches production ciphertext?	Detects a static state with $k = t - 1$ ready shares	Cost per check ($n = 32$, ffdhe2048, measured)
Liveness heartbeat	No	No	No: a node that lost its share still answers	1 message per node; no group operations
Share registration (Feldman/PVSS check)	Once, at setup	No	No: says nothing about later share loss	One-time $O(nt)$ exponentiations at epoch setup
Full-committee canary probe	Yes	No (canary)	Yes, with probability 1	32 proofs + 32 verifications; 8501 ms CPU; 36,991 B
KEYSTONE sampled canary ($s = q = 8$)	Yes, for the sampled members	No (canary)	Yes, with probability 0.9807 (Eq. 2)	8 proofs + 8 verifications; 2125 ms CPU; 9,415 B
Decrypt a real record as a test	Yes	Yes: exposes a production record	Yes	t verifications plus a real disclosure; excluded by design

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